

PREM SHRIKANT GHOSH

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PROFESSIONAL SUMMARY

Computer Science and Information Technology graduate with hands-on software engineering experience and a strong foundation in C++, data structures, debugging, full-stack development, REST APIs, databases, and application optimization. Experienced in identifying functional and performance issues, debugging applications, developing backend services, and building end-to-end software systems.

Strong interest in Artificial Intelligence, Generative AI, high-performance computing, AI inference optimization, and efficient deployment of machine learning models. Currently expanding knowledge of PyTorch, LLM inference, quantization, model optimization, GPU computing, and inference runtimes. Strong analytical and problem-solving abilities with experience working across development, testing, deployment, and technical documentation.

TECHNICAL SKILLS

Programming Languages: C++, C, JavaScript

Core Computer Science: Data Structures, Algorithms, Object-Oriented Programming, Problem Solving, Debugging

AI / ML: Generative AI Fundamentals, Machine Learning Fundamentals, LLM Concepts, AI Inference Concepts

Currently Exploring: PyTorch, llama.cpp, LLM Quantization, Model Inference Optimization, CUDA & GPU Computing

Frontend: React.js, Next.js, HTML5, CSS3

Backend: Node.js, Express.js, REST APIs

Databases: MySQL, MongoDB, PostgreSQL, Firebase

Development Tools: Git, GitHub, Postman, VS Code, Cursor AI, Claude AI, ChatGPT

Software Engineering: REST APIs, JWT Authentication, CRUD, Testing, Debugging, Performance Analysis, Technical Documentation

PROFESSIONAL EXPERIENCE

Web Trainee & Digital Marketing Intern

BrandBase Capsule Pvt. Ltd. | July 2026 – Present

- Maintain and improve production web applications and company digital infrastructure.
- Identify and resolve website functionality, deployment, responsiveness, and performance issues.
- Assist with application deployment, website optimization, SEO, and production updates.
- Collaborate on client software projects and support application bring-up and deployment activities.
- Analyze application behavior across devices and environments and implement improvements.
- Prepare and manage technical and digital content for company platforms.

Software Engineer Intern

Gohil Tech Empire Pvt. Ltd. | May 2026 – June 2026

- Debugged functional and UI/UX issues in the Way To Event software platform.
- Performed application testing and systematic debugging to improve software stability.
- Investigated implementation issues and contributed fixes across application components.
- Worked on software validation and documented implementation findings.
- Authored technical content covering visionOS and technologies introduced at Apple WWDC.
- Prepared technical documentation and implementation reports to support engineering activities.

Full Stack Developer Intern

ElevanceSkills

- Developed responsive software applications using React.js, JavaScript, HTML5, and CSS3.
- Designed and implemented backend APIs using Node.js and Express.js.

- Developed authentication systems and RESTful application services.
- Integrated databases and implemented CRUD-based application functionality.
- Debugged frontend/backend integration issues and improved application reliability.
- Used Git and GitHub for version control, source management, and development collaboration.

PROJECTS

FastCab – Smart Web-Based Cab Booking System

Technologies: React.js, Node.js, Express.js, MySQL

- Designed and developed an end-to-end cab booking system supporting user and driver workflows.
- Developed RESTful backend APIs for booking and ride-management operations.
- Designed MySQL-backed data management for users, drivers, and bookings.
- Implemented authentication, authorization, ride booking, and role-based application functionality.
- Debugged frontend/backend integration and database-related issues throughout development.
- Applied modular software design and CRUD operations to improve maintainability.

Student Management System – Multi-Institute SaaS

Technologies: React.js, Node.js, Express.js, PostgreSQL, Prisma ORM

- Developed a multi-institute software platform with secure institute registration and administration.
- Implemented student CRUD operations, search, filtering, image handling, analytics, and activity logging.
- Designed role-based access control and JWT authentication.
- Developed REST APIs and integrated PostgreSQL using Prisma ORM.
- Debugged application and API integration issues across the software stack.
- Designed the system with scalability and maintainability in mind.

YouTube Clone

Technologies: React.js, Next.js, Node.js, MongoDB

- Developed a responsive video-streaming web application inspired by YouTube.

- Built reusable React components to improve software maintainability.
- Implemented video searching and category-based filtering.
- Integrated backend APIs and MongoDB for application data management.
- Optimized the interface for different screen sizes and application environments.

AREAS OF TECHNICAL INTEREST

- Large Language Models and Generative AI
- AI Model Inference and Deployment
- C++ Systems Programming
- GPU Computing and CUDA
- Model Quantization and Optimization
- AI Inference Runtimes
- Memory and Performance Optimization
- High-Performance Computing
- PyTorch and llama.cpp
- Edge AI and Local Model Deployment

EDUCATION

Bachelor of Science in Information Technology

University of Mumbai | 2023 – 2026

CGPI: 9.53

Strong academic foundation in programming, software development, databases, computer science fundamentals, and information technology.

Higher Secondary Certificate (HSC)

Maharashtra State Board

Percentage: 77.83%

Secondary School Certificate (SSC)

Maharashtra State Board

Percentage: 76.80%

CERTIFICATIONS

- Full Stack Web Development – ElevanceSkills
- Introduction to Generative AI
- Claude 101
- Claude Code 101
- Fundamentals of Computers

ACHIEVEMENTS

- Ranked **1st in the Information Technology Department** for academic excellence.
- Completed internships spanning Software Engineering, Full Stack Development, and Web Development.
- Developed multiple end-to-end software applications using modern frontend, backend, and database technologies.
- Gained practical experience in debugging, testing, deployment, API development, and technical documentation.
- Continuously expanding technical knowledge toward AI/ML, Generative AI, C++ systems programming, and high-performance model inference.

SOFT SKILLS

- Analytical Problem Solving
- Software Debugging
- Technical Documentation
- Performance-Oriented Thinking
- Team Collaboration
- Communication
- Adaptability
- Continuous Learning